INCH-POUND

A-A-52027
26 February 1991

COMMERCIAL ITEM DESCRIPTION

PUMPING UNIT, FRAME MOUNTED, 1-1/2 INCH, ELECTRIC MOTOR DRIVEN, 35 GPM AT 25 FEET TOTAL HEAD

The General Services Administration has authorized the use of this commercial item description in preference to MIL-P-53065.

This item description covers an electric motor driven pumping unit and the associated hoses and fittings. The pump has a capacity of 35 GPM at 25 feet total head.

Salient characteristics:

General. When specific requirements are not stated herein, all items listed as standard equipment in the manufacturers published specifications, brochures, and catalogs which are normally furnished to the commercial customer as standard equipment shall be furnished. In addition, the pumping units shall be equipped with optional equipment and accessories as required to comply with the performance requirements specified herein. Optional equipment for the purpose of this item description may be defined as equipment not standard and not normally furnished with the pumping units but which is available to the commercial customer on an optional basis. Any additional equipment or accessories needed to satisfy military requirements not stated herein shall be specified in the contract or order. The additional equipment shall be constructed, assembled and equipped with all components (whether or not specified herein) necessary to enable the unit to function reliably and efficiently in sustained operation. The pumping units shall conform to all Federal laws and regulations applicable for this type equipment which are in effect on date of contract award.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and other data which may improve this document should be sent by letter to: Commander, US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 4320

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Design and performance.

Pump. The pump shall be self priming and either the centrifugal or the positive displacement type. The pump case and all the internal working parts which contact the pumped fluid shall be of a nonsparking material compatible with diesel fuel, kerosene, and leaded gasoline. The pump casing shall incorporate a drain receptacle and have the suction and discharge ports identified by casts in the casing. A direction arrow shall be cast in the centrifugal pump casing to indicate the direction of impeller rotation. The pump shall be equipped with self-lubricating mechanical seals. The pump shall be furnished with a suction strainer with a maximum of 40-mesh openings. The pump shall be capable of transferring from 35 to 50 GPM of diesel fuel in the viscosity range of 1.3 to 5.0 centistokes at 20 ^{o}C at 20 feet of head. The pump shall prime within two minutes on a 5-foot static suction lift and shall deliver fuel at its rated capacity when positioned at an angle up to 15 degrees from the horizontal. The positive displacement pump shall be provided with a bypass system to avoid overpressurization of the hose assemblies when at a no-flow condition.

Electric motor. The motor shall conform to NEMA MG-1, be direct current, 24 volt, totally enclosed, non-ventilated, and water-proof. The motor shall be explosion-proof and conform to UL 674 for class I, groups C and D and class II, groups F and G. The electric motor shall be equipped with a thermostat to sense approaching overheat conditions and shut off the motor before damage occurs. The electric motor shall be provided with short circuit and overload protection. The electrical circuitry, including all components and connections that are not hermetically sealed or inherently resistant, shall be protected from the effects of moisture and fungus growth.

Protective frame. The unit shall be housed in a protective frame. The complete assembly shall be a maximum size of 28 inches long, 18 inches wide, and 18 inches high, with a maximum weight of 80 pounds. The frame shall allow access for operation and maintenance and be of sufficient strength to allow the unit to be dropped from a height of four feet. The frame shall contain provisions for storing the slave cable and have handles or grasp areas to allow for two-soldier transportability. The frame shall be labeled “ TWO SOLDIER CARRY" in 1-1/2 (±1/8) inch high characters when the frame width permits, otherwise, the characters shall be as high as the frame width allows. The character color shall contrast with the background color.

Accessories.

Grounding assembly. A grounding assembly shall be provided with each unit. The grounding assembly shall include 75 ±1 inches of no. 10 wire connected to two "plier type" ground terminals and a copper clad steel grounding rod 1/2 inch diameter by 60 inches long. The pumping unit shall incorporate a non-painted grounding lug to accommodate a ground terminal.

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Slave cable. The pumping unit shall be equipped with a minimum 15-foot slave cable of sufficient size to safely carry the power to operate the unit at rated conditions. The cable shall be oil resistant and incorporate a slave connection conforming to MIL-C-62122.

Hose. The pumping unit shall be furnished with two 25-foot long suction hoses conforming to MIL-H-370, type I, size 5 with class I, style A fittings. The pumping unit shall also be provided with two 25-foot, long discharge hoses conforming to MIL-H-11588, type III with fittings as specified above. The ends of all wires within the suction and discharge hoses shall be securely bonded to the coupling half with not less than 0.5 inch of each wire in contact with the couplings.

Nozzle. The pump unit shall be furnished with a nozzle. The nozzle shall be a 1-1/2 inch, fuel and oil servicing, nonautomatic shutoff type. The nozzle shall be furnished with a straight spout for use with leaded gasoline. The nozzle shall be adaptable to the discharge hose specified above.

Label. A label stating "Test all fuels of unknown quality before pumping into equipment fuel tanks" shall be placed on the pumping unit in a location which allows the largest character size to be used and can be easily seen.

Color. Unless otherwise specified in the contract or order, the color of the pumping unit shall be Tan 686.

Certification. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer’s own drawings, specifications, standards and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under provisions of the contract.

Regulatory requirements. In accordance with Federal Acquisition Regulation, section 23.403, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or order.

Safety. All shafts or rotating parts of the pumping unit shall be guarded when such parts are exposed to contact by personnel. External surfaces shall be free of sharp edges and corners except where necessary and not detrimental to safety. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

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Quality assurance. The pumps shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

1. Pump design and performance not as specified.
2. Electric motor design and performance not as specified.
3. Protective frame design and performance not as specified.
4. Accessories not as specified.
5. Preservation, packing, and marking not as specified in the contract or order.
6. Safety not as specified.
7. Color not as specified.

Notes: The procuring agency should specify the preferred options permitted herein and include the following information in procurement documents:

1. Title, number, and date of this item description.
2. Any additional equipment or accessories needed to satisfy military requirements.
3. Level of preservation, packing, and marking required.
4. Color, if other than Tan 686.
5. When this commercial item description is used for procurement, the certification clause must appear in the solicitation.
6. The following list of suppliers:

Dismas Pumps
P 0 Box 80008
Billings, MT 59108-0008
Phone (800) 874-8976

CH & E Manufacturing Company, Inc.
3849 N. Palmer Street
Milwaukee, WI 53212
Phone (414) 964-3400

The Gorman-Rupp Company
P 0 Box 1217
Mansfield, OH 44901
Phone (419) 755-1011

Burks Pumps, Inc.
420 E. Third Street
Piqua, OH 45356
Phone (513) 773-2442

Copies of MIL-H-370, MIL-H-11588, and MIL-C-62122 are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

Copies of NEMA, MG-1 is available from the National Electrical Manufacturers Association, 2101 L Street, NW, Suite 300, Washington, DC 20037.

Copies of UL 674 is available from the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062

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Preparing activity:
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